

Landau's bounds on the early zeros

necessary density, local deficit, no constant A

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Abstract

Landau's necessary conditions say that a sampling set for $PW_{L/2}$ must have lower density $D^- \geq L/(2\pi)$. Globally the zeros satisfy this (the density tends to infinity). Locally, on the desert and on $[0, 2\pi\mu]$, they do not: the Landau dimension of functions concentrated there exceeds the number of zeros by 4 to 90, depending on the window. That deficit is why c_L is exponentially small. It does not produce the constant A in the sampling inequality — Landau is a necessary count of modes, not a Slepian eigenvalue. Beurling's sufficiency is global and too coarse to see the desert.

1 What Landau proves

For the Paley–Wiener space PW_Ω of type Ω : if Λ is sampling then $D^-(\Lambda) \geq \Omega/\pi$; if Λ is interpolating then $D^+(\Lambda) \leq \Omega/\pi$. Here $\Omega = L/2$, so the critical density is

$$\rho = \frac{L}{2\pi} = \frac{1}{2\pi/L},$$

the reciprocal of the Nyquist spacing. The proof is spectral: the concentration operator $\chi_I P_\Omega \chi_I$ has

$$\#\{\text{eigenvalues close to } 1\} = \frac{\Omega}{\pi} |I| + o(|I|) = \frac{L}{2\pi} |I| + o(|I|)$$

as $|I| \rightarrow \infty$ (Landau 1967). Those eigenfunctions live on I and are invisible to $\Lambda \setminus I$. If $\#(\Lambda \cap I)$ is smaller than that dimension, Λ cannot sample PW_Ω .

2 Global versus local

The Riemann zeros (and those of a fixed primitive L -function) have $N(T) \sim (T/2\pi) \log(T/2\pi)$. On $\mathbb{R}$ one has $D^- = \infty > \rho$ for every finite L . Beurling's sufficiency, applied to a separated subset as in *sampling-floor* Theorem 1, therefore gives $c_L > 0$ under RH. The bound does not see any desert: D^- is a lim sup of counts in long intervals, and long intervals are well filled.

The desert is a *short* interval. Landau's count on $I = [-\gamma_1, \gamma_1]$ is finite and sharp enough to be used:

$$\dim\{\text{functions concentrated on } [-\gamma_1, \gamma_1]\} \approx \frac{L\gamma_1}{\pi}.$$

One even pair $\{\pm\gamma_1\}$ is two points. The local density $1/\gamma_1$ against ρ is 0.16–0.40 for ζ on the windows below.

3 Measured deficit

Number of positive zeros up to T versus Landau dimension LT/π (even window, $I = [-T, T]$).

set	μ	T	need LT/π	have	deficit	D_{loc}/ρ
ζ	3	γ_1	4.9	1	3.9	0.40
ζ	11	γ_1	10.8	1	9.8	0.19
ζ	11	$2\pi\mu$	52.8	16	36.8	0.61
ζ	16	$2\pi\mu$	88.7	29	59.7	0.65
χ_5	30	γ_1	7.2	1	6.2	0.28
χ_5	30	150	161	89	72	1.11
χ_{17}	11	γ_1	2.9	1	1.9	0.70

On the desert the set is always subcritical (ratio 0.2–0.7). On $[0, 2\pi\mu]$ it climbs toward criticality and crosses it for χ_5 once the cache is long enough — that crossing is why a finite list of zeros can look “dense enough” while c_L is still $e^{-s\mu}$: the *first* LT/π modes, not the last, set the floor.

4 What the bound does not give

Landau does not estimate the gap $1 - \lambda$ of the concentration operator, only how many eigenvalues sit near 1. The size of c_L is that gap for the ground state of the Gram of Λ , which is a Slepian problem on a multi-band set (the complement of the early gaps). Replacing the count “deficit = 4” by “ $-\ln c_L = 16.7$ ” is exactly the passage this note does not make — it is *desert-slepian*, and the factor is 2.5–6.

Beurling’s $D^- > \rho$ is sufficient after thinning, and ignores the same short interval. A quantitative Landau inequality with an explicit A in terms of $D^- - \rho$ exists only in asymptotic form for long sets; it does not cover a hole of length γ_1 followed by twenty sparse points.

Status

Landau names the obstruction: on every window the early zeros are a locally subcritical set, by 4 to 90 modes. That is the necessary half of the sampling theorem already used qualitatively in Theorem 1. It is not a formula for c_L . The sufficient half with a constant remains the Slepian of the early set as a whole. Not RH.